

Rotational Motion

SHORT NOTES

Moment of inertia (I)

Moment of Inertia is a scalar (positive quantity).

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots$$

$$= I_1 + I_2 + I_3 + \dots$$

For a single particle

$$I = mr^2$$

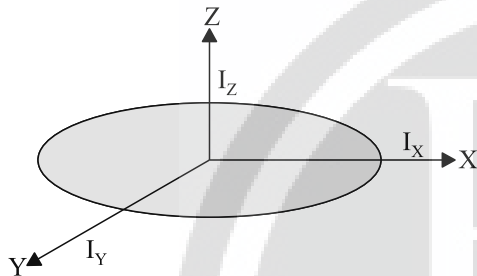
For a continuous object

$$I = \int dI = r^2 \int dm$$

Perpendicular Axis Theorem

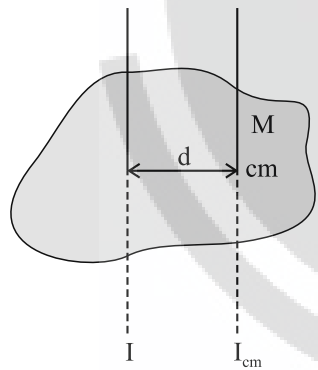
Only applicable to plane lamina (that means for 2-D objects only)

$$I_Z = I_X + I_Y \text{ (when object is in } x-y \text{ plane).}$$



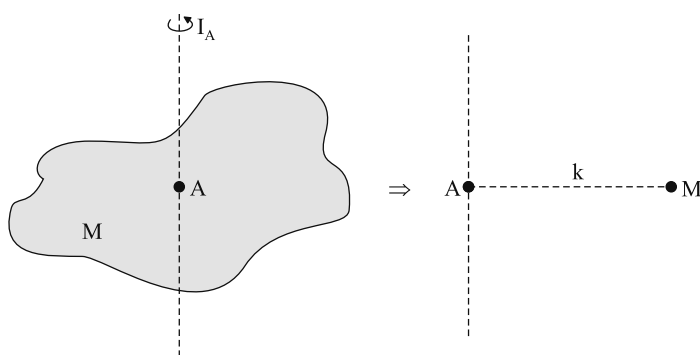
Parallel Axis Theorem

(Applicable to any type of object):



$$I = I_{cm} + Md^2$$

Radius of gyration (k)



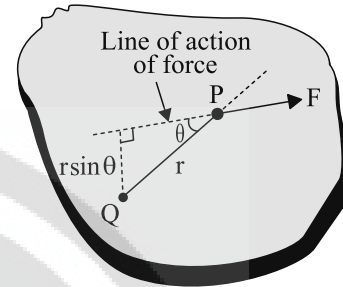
$$I_A = Mk^2$$

$$k = \sqrt{\frac{I_A}{M}}$$

Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = rF \sin \theta$$



Rotational equilibrium

For translational equilibrium.

$$\sum F_x = 0$$

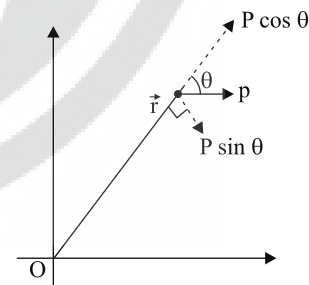
$$\text{and } \sum F_y = 0$$

The condition of rotational equilibrium is

$$\sum \vec{\tau} = 0$$

Angular momentum ($\vec{L}$)

Angular Momentum of a Particle About a Point



$$\vec{L} = \vec{r} \times \vec{P}$$

$$L = rP \sin \theta$$

$$|L| = r_{\perp} \times P$$

$$|L| = P_{\perp} \times r$$

Conservation of Angular Momentum

Angular momentum of a particle or a system remains constant if $\tau_{\text{ext}} = 0$ about that point or axis of rotation.

$$L_i = L_f \Rightarrow I_i \omega_i = I_f \omega_f$$

Relation Between Torque and Angular Momentum

$$\tau = \frac{dL}{dt} = I\alpha$$

Torque is change in angular momentum

Impulse of Torque

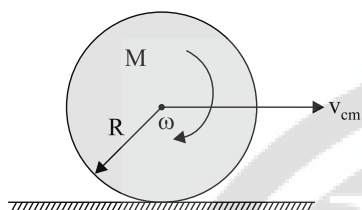
$$\int \tau dt = \Delta J$$

ΔJ = Change in angular momentum or angular impulse.

Rolling Motion

$$\text{Total kinetic energy} = \frac{1}{2} M v_{\text{CM}}^2 + \frac{1}{2} I_{\text{CM}} \omega^2$$

$$\text{Total angular momentum} = M v_{\text{CM}} R + I_{\text{CM}} \omega$$



Pure Rolling (or Rolling without Slipping) on Stationary Surface

Condition: $v_{\text{CM}} = R\omega$

In accelerated motion, $a_{\text{CM}} = R\alpha$

If $v_{\text{CM}} > R\omega$ then rolling with forward slipping.

If $v_{\text{CM}} < R\omega$ then rolling with backward slipping.

Total kinetic energy in pure rolling

$$K_{\text{total}} = \frac{1}{2} M v_{\text{CM}}^2 + \frac{1}{2} (Mk^2) \left(\frac{v_{\text{CM}}^2}{R^2} \right) = \frac{1}{2} M v_{\text{CM}}^2 \left(1 + \frac{k^2}{R^2} \right)$$

Pure rolling motion on an inclined plane

$$\text{Acceleration } a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}}$$

$$\text{Minimum frictional coefficient } \mu_{\text{min}} = \frac{\tan \theta}{1 + \frac{R^2}{k^2}}$$

$$v = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}} = \sqrt{\frac{2gl \sin \theta}{1 + \frac{k^2}{R^2}}} \quad (\text{since } h = l \sin \theta)$$

$$t = \sqrt{\frac{2l(1 + k^2/R^2)}{g \sin \theta}}$$

$$= \sqrt{\frac{2h(1 + k^2/R^2)}{g \sin^2 \theta}} \quad (\text{since } h = l \sin \theta)$$

Angular momentum about axis O $\vec{L}_O = \vec{L}$ about C.M. + $\vec{L}$ of C.M. about O

$$\vec{L}_O = I_{\text{CM}} \vec{\omega} + \vec{r}_{\text{CM}} \times M \vec{v}_{\text{CM}}$$

